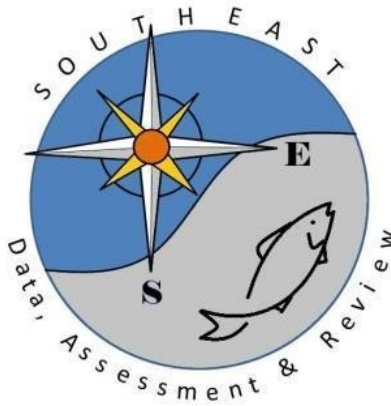




SEDAR

Southeast Data, Assessment, and Review

SEDAR 102

ASMFC Atlantic Menhaden and Ecological Reference
Points

Research Recommendations

October 2025

SEDAR
4055 Faber Place Drive, Suite 201
North Charleston, SC 29405

The panel strongly supports the proposed research recommendations, emphasizing three key priorities:

1. Data Collection: Continue and expand collection of population, life history, and diet data across all ecosystem components
2. Spatial Assessment: Hold a stakeholder workshop to determine clear objectives for a potential spatial ERP assessment before model development
3. Assessment Coordination: Ensure adequate time between single-species assessments and ERP work for proper model development and analysis

ToR 9. Review the research, data collection, and assessment methodology recommendations provided by the TC and make any additional recommendations warranted. Clearly prioritize the activities needed to inform and maintain the current assessment, and provide recommendations to improve the reliability of future assessments.

The review panel finds that the research recommendations proposed by the ERP WG for future development were reasonable, including those that have been carried over from the previous ERP benchmark. The division of short term and long term data and modeling needs in the research recommendations is extremely useful. It should be noted that the list may be longer than can be achieved, and therefore prioritizing is important. Here we suggest some priorities.

The review panel strongly supports continued and expanded seasonal, spatial, and taxonomic collection of population abundance, biomass, life history, condition, and diet data for both managed fish species, including menhaden, and functional groups throughout the ecosystem from primary producers through zooplankton and benthos to apex fish, bird, and marine mammal predators. Existing fishery independent surveys and fishery dependent data collection must be maintained both for stock assessment and for ecosystem assessment. In addition to stock assessment data, low trophic level productivity and diet data are essential to ERP model function. Improved data on prey size or age from diet collections is recommended to apportion diet across age stanzas of prey. Investigation of other existing datasets such as the Hudson River Biological Monitoring Program (HRBMP) data (1974 to 2018 for menhaden and striped bass) is also recommended.

The review panel strongly supports holding a workshop with managers and stakeholders similar to the 2015 EMO workshop to determine clear objectives for a potential spatial ERP assessment prior to any spatial model development. This workshop should also review data requirements for assessment at different spatial scales alongside existing data to ensure that expectations for spatial assessment are realistic and achievable. In particular it is important to be clear on the degree of spatial resolution possible within a proposed model, and evaluate if this would meet the wishes of the stakeholders and managers. Even if expansion of data collection as recommended above is feasible, historical data could limit the effectiveness of spatial models. In addition to considering multiple existing spatial modeling frameworks for their ability to

integrate existing data and achieve management objectives, the workshop should also consider whether the objectives could be achieved with simpler non-spatial models at different scales, e.g., a model for Chesapeake Bay.

The review panel strongly supports the recommendation of the scientific group regarding assessment coordination and planning for an appropriate amount of time between the single species assessment data becoming available and the deadline for the ERP work, both for management use and especially for future benchmarks. Maintaining the current ERP assessment method requires updating and revising the NWACS model as new data and assessments become available (especially the revised MRIP recreational data and the forthcoming striped bass benchmark). However, continuous updates to the NWACS-MICE would be an inefficient use of limited modeling resources. The panel recommends conducting research on general methods for evaluating parameter uncertainty and sensitivity of ecosystem models in between benchmarks, which should be scheduled well after updated assessments for ERP focal species are completed (see ToR 10).

The review panel highlights the following recommendation for future ERP benchmarks. Once a base case model is developed, the ERP WG should focus on creating a suite of different model formulations which give “plausible” model outputs. These could be generated by making changes to the selected base run. Such a suite of models could then be used to characterize and compare the key uncertainties in the model. Once there is such a suite of models, one possibility would be to consider ensemble model averaging in future management advice. However, the suite would have clear utility in investigating uncertainty regardless of any formal ensemble modelling.

The review panel also suggests these specific research recommendations:

Continue to code and fully document model input data processing, balancing, and fitting protocols along with acceptable ranges for model diagnostics and structured sensitivity runs with the goal of fully repeatable and transparent methods that can be handed off from research teams to agency staff charged with operational assessment.

Evaluate NWACS-MICE sensitivity to initial biomass and productivity – how robust are results to revised assessment total biomass estimates and changing assumptions regarding single species M? Existing input data uncertainty methods such as Ecosampler or the R-based ecosense could be investigated for this process. A hybrid approach investigating the subset of most sensitive parameters could be implemented in alternative software such as Rpath (Lucey et al 2020) or EcoState (Thorson et al 2024) for more efficient sensitivity analysis of static food web and dynamic parameters, respectively.

Investigate NWACS-MICE model sensitivity to prey switching parameters in the absence of data, and or evaluate food habits data along with ecosystem sampling data to detect the extent to which prey switching has been observed.

Evaluate NWACS-MICE behavior with long term egg forcing using recruitment deviations for species in addition to Atlantic herring and determine whether this allows estimation of predator prey parameters away from bounds.

Develop clear objectives for NWACS-Full. If the models are to be used together for future management advice, build out NWACS-Full directly from NWACS-MICE and fit to the same time series in both models to facilitate direct comparisons between results. This may necessitate staggered model development to ensure that the final NWACS-MICE base model informs the NWACS-Full model prior to NWACS-Full fitting, but would have the additional benefit of simplifying the complex task of tuning the NWACS-Full model. The model selection process and tuning protocol should be well documented to ensure transparency and consistency

Collect age data from the menhaden fishery independent surveys to inform M (e.g., are ages up to 10 actually observed? What are the current oldest ages?).

Finally, the review panel supports the development of an MSE to identify harvest strategies maximizing the likelihood of achieving the defined ecosystem management objectives. The MSE would be most effective if it is based on coordinated management objectives between menhaden and their predator species across management Boards, which require development.